

From Difference to Memory in a Minimal Non-Neural Carrier

How contact can leave a trace in the same substrate that later conducts repair

Amir Tlinov Independent research 20 July 2026 materials & source two-page PDF

A STEP BEFORE THE FINISHED OBJECT

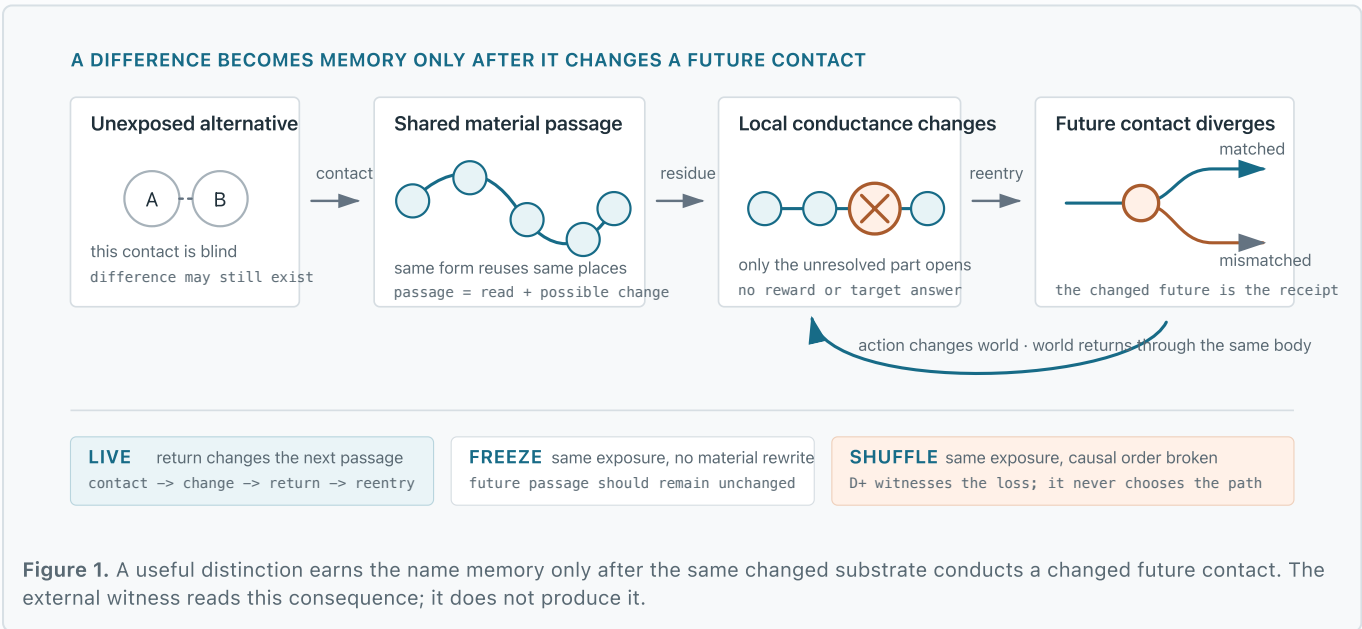
Physics, biology, and computation usually begin with named things—particles, cells, networks, agents—and then describe relations among them. This project tries a narrower reversal. Rather than declare that *everything is difference*, it begins one step earlier than the finished object: with alternatives in a substrate, and with the contacts through which an alternative can matter to what happens next.

Difference is not offered as a replacement for physics. It is a restraint against letting coordinates, particles, graphs, or readouts silently become the process being described. Levin's work on basal cognition motivates a related move across scale: ask what a system can sense, retain, pursue, and repair before deciding which disciplinary category owns the phenomenon.

CONTACT MAKES A DIFFERENCE ACCESSIBLE

A difference may exist without being exposed by one contact. A blind probe does not establish equivalence across every possible continuation. Conversely, measurement is itself physical contact: a registered outcome belongs to the coupled history of system and apparatus, not to a view from outside. This does not solve quantum measurement; it only refuses to treat an observed pattern as a photograph of untouched alternatives.

Operationally, two retained histories count as distinguishable only if replacing one with the other changes some admissible later contact. The formal quantity $D+$ witnesses the strongest such future difference. It is not an organ of the carrier: it never chooses a route, creates a memory, computes a geometry, or tells the substrate what to do.



THE CARRIER HYPOTHESIS

Contact must change the same substrate that later conducts a changed continuation.

1. A compatible closure reuses the same places and later acts as one whole.
2. Only an unresolved real passage opens new conductance.
3. Action must change the world before return can be earned.
4. Reentry must change a later contact and survive restart.
5. freeze / blocked / shuffle must break that exact gain.

1 loop

contact → return →
reentry
changed future is the
receipt

WHAT EXISTS NOW

A new scale is not assigned: it appears only when a closure later enters contact as one whole. The program has passed through two computational forms. The first makes history-dependent repair reproducible under intervention. The second moves memory from an outcome table into the passage itself. Neither is presented as biology or fundamental physics; together they expose the precise missing owner—a material law by which the substrate itself chooses and changes the path.

1 • Public bounded assay

relational-closure-carrier uses explicit local handles and a prepared closure affordance. A/B histories later repair differently after temporary-world deletion, fresh-process restart, common washout, and the same renewed injury. Relation-preserving address relabeling preserves the effect; placement shuffle erases it. This proves causal relational history in the prepared carrier—not self-born topology, boundary, or goals.

2 • Passage becomes persistent

The later one to two PC carrier is one writable mmap body with a local Metal law. Three events reuse the same sixteen conducting relations; a mismatched world return changes their common R^* , changes the next passage, and survives reopen. Severed conductance can regrow from surviving context. Yet programmed ORef relations and the computer still route the material. Its frozen public proof bundle is not yet released.

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public explicit-topology
assay tests pass

2,058

matched events preserved
by hostile relabeling

1 R^* / 3

one shared relation across
three PC events

reopen

changed future remains
after restart

Why regeneration is the bridge

A planarian fragment is not a smaller copy of a completed animal. A cut leaves not an empty blueprint slot but a living organization with an unresolved continuation: its future contacts no longer match the stable response class maintained by the whole. Reentry may recover familiar anatomy, stabilize another morphology, let fragments become new wholes, or end in collapse. Regeneration is a new closure constrained by surviving physiological history—not insertion of a stored part. This is an interpretation, not a molecular explanation.

The physical test that can fail

A random memristive nanowire network would let conductance, heat, and hysteresis alter later physical paths. Parameters are fixed on A and predict held-out B. After damage and reentry, live must change a hidden next contact more than freeze, blocked, and shuffle. No held-out advantage, no survival after power loss, or an externally chosen route defeats the carrier hypothesis.

QUESTION FOR BIOLOGY

Given two tissues matched in present anatomy and immediate behavior but differing only in causal repair history, **what perturbation would best distinguish a distributed homeostatic target from ordinary path-dependent hysteresis**—without supplying the desired anatomy from outside?

SCOPE & REFERENCES

1. **Supported:** retained history can survive restart and renewed injury, remain invariant under irrelevant relabeling, and change a later passage in engineered carriers. **Open:** autonomous closure and scale formation, boundaries, goals, biological relevance, basal cognition, carrier-owned physical routing, and any claims about matter, space, light, or gravity.
2. Levin (2022), *Technological Approach to Mind Everywhere*; Durant et al. (2017), long-term editing of regenerative anatomy; Emmons-Bell et al. (2015), bioelectric induction of alternative planarian head anatomies.
3. Manicka & Levin (2019), somatic computation in non-neural bioelectric networks; Levin (2023), bioelectric networks across scales; Schlosshauer (2005), decoherence and quantum measurement.